

AI-Powered Adaptive Robotic Arm for Enhancing Elderly Independence and Quality of Life

The global aging population presents significant challenges in maintaining independence and quality of life for older adults. To address this, we propose developing an AI-powered assistive robotic arm specifically designed for the elderly. This lightweight, user-friendly device will learn from users' habits, adapt to individual needs, and assist with daily tasks, potentially integrating with smart home systems to enhance independence and well-being.

Research Question: To what extent can an AI-driven, habit-learning robotic arm improve the daily task performance and autonomy of elderly individuals compared to traditional assistive devices?

Background

The global population aged 60 and over is projected to double from 1 billion in 2020 to 2.1 billion by 2050. This demographic shift presents significant challenges in elderly care, with healthcare spending for the elderly expected to increase by 20% by 2030 [1]. Traditional assistive devices often lack adaptability and personalization, limiting their effectiveness for users with cognitive decline or mobility limitations. AI-powered solutions, such as robotic arms, offer personalized interaction, task adaptation, and integration with smart home technology [2],[3]. This research, aligned with the Fulton Schools of Engineering's 'health' theme, aims to develop an innovative AI-powered assistive robotic arm to enhance elderly independence and quality of life. The potential impact extends beyond individual users, potentially reducing caregiver burden and healthcare system strain.

Hypothesis

We hypothesize that an AI-powered, habit-learning robotic arm will significantly improve the daily task performance and autonomy of elderly individuals compared to traditional assistive devices. By adapting to users' unique needs and integrating with smart home systems, this device will enhance independence and reduce the reliance on caregivers.

- **Enhanced Personalization and Adaptability:** Current assistive devices lack the ability to adapt to individual user needs, offering only static support. Our research will enable the development of AI-powered robotic arms that learn and adjust to users' habits, providing highly personalized and adaptive assistance for daily tasks.
- **Promoting Independent Living:** The elderly often rely heavily on caregivers due to the limitations of traditional devices. By introducing a habit-learning, AI-driven robotic arm, our research will empower users to perform tasks independently, significantly reducing reliance on caregivers and fostering greater autonomy [4].
- **Seamless Smart Home Integration:** Traditional assistive technologies do not effectively integrate with smart home systems, limiting their usability. Our research will bridge this gap, enabling the robotic arm to work in harmony with smart home devices, offering comprehensive and intuitive support for elderly individuals in their daily routines.
- **Alleviating Healthcare System Strain:** As the elderly population grows, healthcare systems face increasing pressure due to the high demand for caregiving and support. By extending elderly autonomy through adaptive, AI-powered assistance, our research will help reduce the burden on healthcare services, potentially delaying the need for full-time care and lowering associated costs.

Methodology

1. Project Scope and Objective:

Given the FURI timeline and resources, we will focus on a simplified task to enhance elderly independence. The primary objective is to assist elderly users in picking objects from the ground and placing them back using a commercially available robotic arm.

2. **Robotic Arm Selection:**
We will procure a simple robotic arm from an online retailer, such as Amazon, that is suitable for basic manipulative tasks and can be programmed for our specific application.
3. **Task Design:**
The initial task will involve the robotic arm assisting elderly users in picking up objects from designated locations and placing them back. This task is chosen for its practical relevance and feasibility within our project timeline.
4. **Object Detection:**
To simplify the object recognition process, we will utilize color-based detection. The robotic arm will be programmed to recognize objects primarily by their color, allowing for easier identification in various environments.
5. **User Interaction:**
The elderly user will demonstrate the task by physically guiding their own arm to indicate where the object is located and where it should be placed. The robotic arm will be programmed to follow the user's movements, ensuring a seamless interaction that leverages the user's input.
6. **Testing and Evaluation:**
We will conduct user trials with elderly participants to evaluate the effectiveness of the robotic arm in assisting with the task. Feedback will be collected to assess ease of use, functionality, and overall satisfaction, allowing us to make necessary adjustments.
7. **Iterative Development:**
Based on user feedback and performance assessments, we will refine the programming and functionality of the robotic arm to enhance its effectiveness and user-friendliness.
8. **Documentation and Reporting:**
Throughout the project, we will document our findings, challenges, and successes, culminating in a comprehensive report detailing the outcomes of our simplified task implementation.

Expected Outcome

Enhanced Independence: Elderly users will gain improved autonomy in performing daily tasks with the robotic arm's assistance [4],[5].

Intuitive Interaction: Users will effectively guide the robotic arm using their own movements, promoting user-friendly engagement.

Successful Object Recognition: The robotic arm will accurately identify and retrieve objects based on color detection.

References

- [1] S. T. Burner, D. R. Waldo, and D. R. McKusick, "National health expenditures projections through 2030," *Health Care Financing Review*, vol. 14, no. 1, pp. 1–29, 1992.
- [2] H. Sekkat, S. Tigani, R. Saadane, and A. Chehri, "Vision-based robotic arm control algorithm using deep reinforcement learning for autonomous objects grasping," *Applied Sciences*, vol. 11, no. 17, p. 7917, Aug. 2021. doi:10.3390/app11177917
- [3] K. Zdravkova, V. Krasniqi, F. Dalipi, and M. Ferati, "Cutting-edge communication and learning assistive technologies for disabled children: An artificial intelligence perspective," *Frontiers in Artificial Intelligence*, vol. 5, Oct. 2022. doi:10.3389/frai.2022.970430
- [4] M. Hudec et al., "Robots for Elderly Care: Review, Multi-Criteria Optimization Model and Qualitative Case Study," *Int J Environ Res Public Health*, vol. 20, no. 9, May 2023, doi: 10.3390/ijerph20095431.
- [5] S. Siddiqui et al., "Artificial Intelligence (AI) and Robotics in Elderly Healthcare," *Cureus*, vol. 15, no. 8, Aug. 2023, doi: 10.7759/cureus.42594.

Budget Request

Item	Price	Discount	Final Price	Price after Tax (8.5%)	Link	Description
HIWONDER MasterPi 5DOF AI Vision Robot Arm	\$309.99	-\$15.50 (5%)	\$294.49	\$319.52	HIWONDER Robot Arm	AI vision robot car powered by Raspberry Pi 4B with 5DOF robot arm, mecanum wheels, and HD camera for diverse AI applications.
Tiny Machine Learning Kit [AKX00028]	\$60.00	N/A	\$60.00	\$65.10	Tiny Machine Learning Kit	Starter bundle with Arduino Nano 33 BLE Sense, OV7675 camera, and custom shield for hands-on TinyML projects.
Total Price (after tax)				\$384.62		

Timeline

Month	Phase	Dates	Tasks
January 2025	Research and Planning	Jan 13 - Jan 24 (Week 1-2)	Conduct literature review on AI-powered robotic arms for elderly assistance
		Jan 27 - Jan 31 (Week 3)	Select and order MasterPi 5DOF AI Vision Robot Arm, and Tiny Machine Learning Kit (for additional AI capabilities) on Amazon
February 2025	Research and Planning	Feb 3 - Feb 7 (Week 4)	Develop detailed project plan and task breakdown
		Feb 10 - Feb 14 (Week 5)	Set up Raspberry Pi 4B for robotic arm control and install AI vision processing libraries.
	Assembly and Implementation	Feb 17 - Feb 21 (Week 6)	Assemble and calibrate the MasterPi robot arm.
		Feb 24 - Feb 28 (Week 7)	Implement basic movement controls and color-based object detection using the HD camera
March 2025	Assembly and Implementation	Mar 3 - Mar 7 (Week 8)	Develop user interaction interface for arm guidance
		Mar 10 - Mar 14 (Week 9)	Conduct initial functionality tests and debug basic operations
	Testing Debugging Improvement	Mar 17 - Mar 21 (Week 10)	Implement iterative improvements based on initial test results
		Mar 24 - Mar 28 (Week 11)	Conduct trials with 5-10 elderly participants to gather feedback on usability and effectiveness.
		Mar 31 - Apr 4 (Week 12)	Analyze user feedback and identify areas for enhancement
		Apr 7 - Apr 11 (Week 13)	Implement final improvements based on user feedback
April 2025	Finalization and Presentation	Apr 14 - Apr 18 (Week 14)	Conduct comprehensive system testing and performance evaluation
		Apr 21 - Apr 25 (Week 15)	Prepare project documentation and final report
		Apr 28 - May 1 (Week 16)	Create and rehearse FURI presentation
		May 2 (Week 17)	Participate in the FURI Symposium and present research findings

Personal Statement

As a sophomore in Robotics Engineering at ASU, I am excited to apply for the Fulton Undergraduate Research Initiative (FURI) program. My passion for microcontroller applications in robotics, coupled with my hands-on experience and desire to make a positive impact, aligns perfectly with FURI's mission to foster innovation and research skills in undergraduate students.

My journey in robotics began in EGR 102, where I led the coding for an Arduino-based rain gauge project, winning recognition for the most creative product. This experience ignited my curiosity about microcontrollers in real-world applications. As a documentation lead in the EPICS Powerpaws: Groceries Shopping project, I've gained insights into how robotics can enhance life quality for individuals with disabilities. Additionally, winning first prize in the Project Humanities hackathon with "Diego's Journey," a game app detecting early signs of mental disabilities in children, demonstrated technology's power in addressing critical social issues.

My research interests focus on exploring microcontroller applications in robotics, particularly:

1. Processing sensor data and controlling motors
2. Implementing decision-making algorithms
3. Developing coordinated motion control systems
4. Exploring advanced microcontroller programming techniques

I'm drawn to projects that can improve the lives of the elderly, inspired by my grandparents' challenges. I envision creating cost-effective, user-friendly robotic solutions to assist seniors in daily activities. I'm also intrigued by robotics' potential in environmental conservation, such as waste separation and water filtration.

Participating in FURI would provide an invaluable opportunity to deepen my technical skills and knowledge in robotics engineering. I'm eager to contribute to cutting-edge research under experienced faculty guidance, particularly Dr. Anoop Grewal, whose interests closely align with mine. Through FURI, I aim to:

1. Develop research skills and contribute to meaningful projects
2. Gain hands-on experience in advanced microcontroller applications
3. Explore the intersection of robotics and social impact
4. Network with like-minded peers and faculty members
5. Prepare for a future career in robotics research and development

I am committed to leveraging this opportunity to not only advance my own skills but also to contribute to research that has the potential to improve lives and address global challenges. The FURI program represents a crucial step in my journey to become an innovative robotics engineer capable of creating technology that makes the world a better place.

Thank you for considering my application. I look forward to contributing to the vibrant research community at ASU through the FURI program.

Phuong Uyen Nguyen (Luna Nguyen)

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Education

Bachelor of Science in Engineering (Engineering Robotics)

Arizona State University, Polytechnic, Arizona

May 2027

4.2/4.0

- Awards/Scholarships: New American University - University Award

Professional Experience

Student Success Aide, School of Manufacturing Systems and Networks (MSN) September 2024 - Present

Arizona State University Polytechnic Campus, Mesa, Arizona

- Scheduled student appointments via the Fulton Schools of Engineering online tool and answered multi-line phones, ensuring efficient communication and seamless appointment management for students and staff.
- Assisted approximately 20 students in a single day by delivering detailed information about the MSN program.
- Managed at least 5 Salesforce cases per day, assisting students by providing clear and thorough email responses on Salesforce.

Project Experience

EPICS, PowerPaws Service Dog Grocery Store Shopping

January 2024 - May 2024

Arizona State University, Tempe, Arizona

Documentation Lead

- Managed project documentation, ensuring timely submission and quality of all deliverables.
- Collaborated on project management, including requirements gathering, task delegation, and progress monitoring.
- Facilitated team communication and understanding, clarifying tasks and addressing concerns to promote effective teamwork.

User-Centered Project_Agriculture Sector, Fulton Schools of Engineering

April 2024

Arizona State University, Mesa, Arizona

- Collaborated in a 4-person team to develop an innovative rain gauge system using Arduino, servo motors, and an IR sensor, winning the "Most Innovative/Creative" award among competing projects.
- Programmed Arduino and wired electrical components, including servo motors and IR sensor, to create a functional prototype for automated daily flushing and irrigation speed control.
- Contributed to project planning, created Gantt chart, and delivered elevator pitch for final showcase.

Hacks for Humanity, Project Humanities, Scottsdale, Arizona

October 2023

- Collaborated with a team of 5 to develop a solution (website and app) for children with mental disabilities.
- Managed time effectively to finish the product, pitch video, and presentation in 36 hours.
- Designed and developed the app named Diego's Journey to aid in early detection of mental disability patterns in young children, earning the First Prize.

Skills

Technical: Microsoft Office Suite (Word, PowerPoint, and Excel), Google Suite (Doc, Sheet, and Slide), Arduino IDE

Languages: Vietnamese (Native), English (fluent)

Arizona State University
Unofficial Transcript

Name: Phuong Uyen Nguyen
Student ID: 1229611684

Print Date:	10/16/2024			Term GPA:	4.00	Term Totals	<u>Attempted</u>	<u>Earned</u>	<u>Points</u>
	Beginning of Undergraduate Record			Cum GPA:	4.00	Cum Totals	37.000	37.000	154.660

END OF TRANSCRIPT

2023 Fall					
Course	Description	Attempted	Earned	Grade	Points
ASU 101-TPS	The ASU Experience	1.000	1.000	A+	4.333
EGR 101	Found of Eng Design Proj I	3.000	3.000	A+	12.999
ENG 107	First-Year Composition	3.000	3.000	A+	12.999
FIN 123	ASU Financial Literacy	1.000	1.000	A	4.000
HSE 101	Human Systems Engineering	3.000	3.000	A+	12.999
MAT 265	Calculus for Engineers I	3.000	3.000	A+	12.999
PHI 105	Intro to Ethics	3.000	3.000	A-	11.001
		<u>Attempted</u>	<u>Earned</u>	<u>Points</u>	
Term GPA:	4.00	Term Totals	17.000	17.000	71.330
Cum GPA:	4.00	Cum Totals	17.000	17.000	71.330

Academic Standing: Good Standing
Term Honor: Dean's List

2024 Spring					
Course	Description	Attempted	Earned	Grade	Points
CHM 113	General Chemistry I	4.000	4.000	A+	17.332
EGR 102	Found of Eng Design Proj II	3.000	3.000	A+	12.999
EGR 104	Crit Inquiry in Engineering	3.000	3.000	A	12.000
ENG 108	First-Year Composition	3.000	3.000	A+	12.999
FSE 104	EPICS Gold I	1.000	1.000	A	4.000
MAT 266	Calculus for Engineers II	3.000	3.000	A	12.000
		<u>Attempted</u>	<u>Earned</u>	<u>Points</u>	
Term GPA:	4.00	Term Totals	17.000	17.000	71.330
Cum GPA:	4.00	Cum Totals	34.000	34.000	142.660

Academic Standing: Good Standing
Term Honor: Dean's List

2024 Fall					
Course	Description	Attempted	Earned	Grade	Points
EGR 201	Use-Inspired Design Project I	3.000	0.000	NR	0.000
EGR 216	EGR Electrical Fundamentals	3.000	0.000	NR	0.000
EGR 218	Materials and Manufact Process	3.000	0.000	NR	0.000
MAT 267	Calculus for Engineers III	3.000	3.000	A	12.000
PHY 121	Univ Physics I: Mechanics	3.000	0.000	NR	0.000
PHY 122	University Physics Lab I	1.000	0.000	NR	0.000
TEM 200	My Technology Venture	3.000	0.000	NR	0.000